A Generic View of Process

Software engineering applies systematic, quantifiable and repeatable approaches to develop high-quality systems efficiently and cost-effectively. Key aspects include fundamental development activities, life cycle models to structure the overall process, and assessment methods to drive continuous improvement. A generic process framework for software engineering defines five framework activities— communication, planning, modeling, construction, and deployment.

Software Engineering – A Layered Technology

Software engineering is a fully layered technology, to develop software we need to go from one layer to another. All the layers are connected and each layer demands the fulfillment of the previous layer.

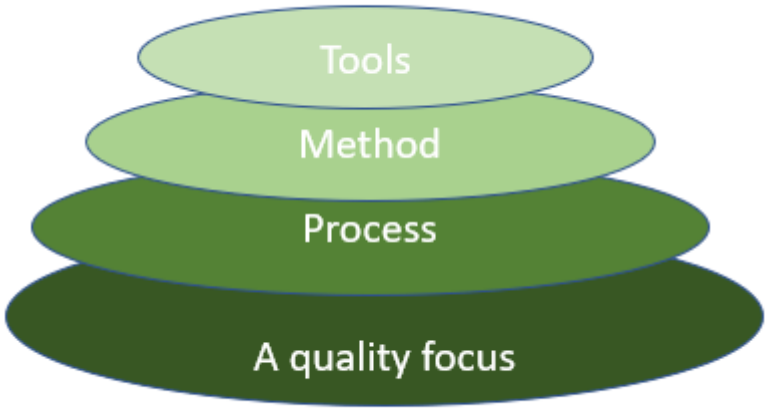
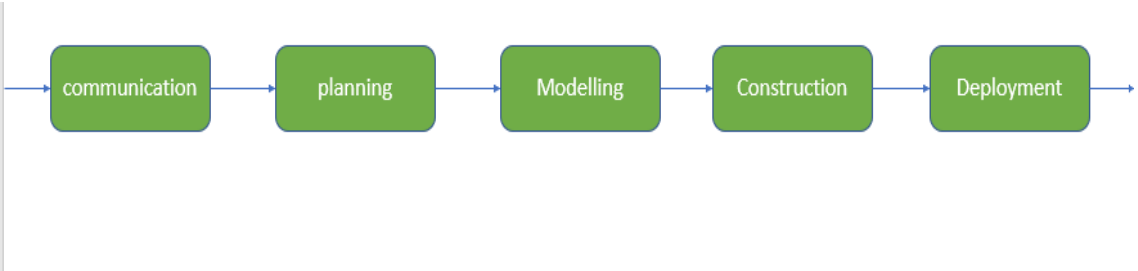


Fig: The diagram shows the layers of software development

Layered technology is divided into four parts:

1.A quality focus: It defines the continuous process improvement principles of software. It provides integrity that means providing security to the software so that data can be accessed by only an authorized person, no outsider can access the data. It also focuses on maintainability and usability.

2. Process: It is the foundation or base layer of software engineering. It is key that binds all the layers together which enables the development of software before the deadline or on time. Process defines a framework that must be established for the effective delivery of software engineering technology. The software process covers all the activities, actions, and tasks required to be carried out for software development.



Process activities are listed below:-

- **Communication:** It is the first and foremost thing for the development of software. Communication is necessary to know the actual demand of the client.
- **Planning:** It basically means drawing a map for reduced the complication of development.
- **Modeling:** In this process, a model is created according to the client for better understanding.
- **Construction:** It includes the coding and testing of the problem.
- **Deployment:-** It includes the delivery of software to the client for evaluation and feedback.

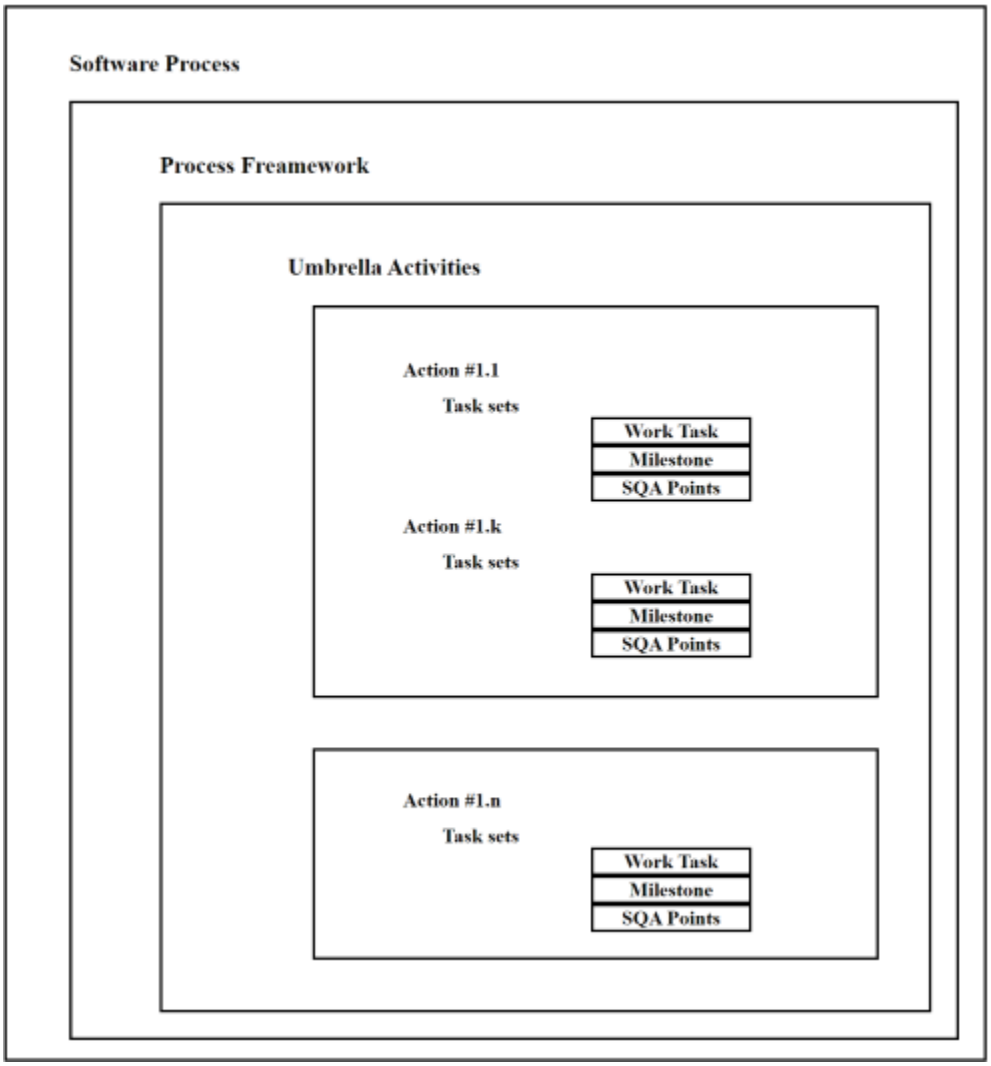
3. Method: During the process of software development the answers to all "how-to-do" questions are given by method. It has the information of all the tasks which includes communication, requirement analysis, design modeling, program construction, testing, and support.

4.Tools: Software engineering tools provide a self-operating system for processes and methods. Tools are integrated which means information created by one tool can be used by another.

Process Framework

Software Process Framework details the steps and chronological order of a process. Since it serves as a foundation for them, it is utilized in most applications. Task sets, umbrella activities, and process framework activities all define the characteristics of the software development process. Software Process includes:

1. **Tasks:** They focus on a small, specific objective.
2. **Action:** It is a set of tasks that produce a major work product.
3. **Activities:** Activities are groups of related tasks and actions for a major objective.



Software Process Framework

Process Framework Activities

The process framework is required for representing common process activities. Five framework activities are described in a process framework for software engineering. Communication, planning, modeling, construction, and deployment are all examples of framework activities. Each engineering action defined by a framework activity comprises a list of needed work outputs, project milestones, and software quality assurance (SQA) points.

1. Communication

By communication, customer requirement gathering is done. Communication with consumers and stakeholders to determine the system's objectives and the software's requirements.

2. Planning

Establish engineering work plan, describes technical risk, lists resources requirements, work produced and defines work schedule.

3. Modeling

Architectural models and design to better understand the problem and to work towards the best solution. The software model is prepared by:

- Analysis of requirements
- Design

4. Construction

Creating code, testing the system, fixing bugs, and confirming that all criteria are met. The software design is mapped into a code by:

- Code generation
- Testing

5. Deployment

In this activity, a complete or non-complete product or software is represented to the customers to evaluate and give feedback. On the basis of their feedback, we modify the product for the supply of better products.

Umbrella Activities

Umbrella Activities are that take place during a software development process for improved project management and tracking.

1. **Software project tracking and control:** This is an activity in which the team can assess progress and take corrective action to maintain the schedule. Take action to keep the project on time by comparing the project's progress against the plan.

2. **Risk management:** The risks that may affect project outcomes or quality can be analyzed. Analyze potential risks that may have an impact on the software product's quality and outcome.

3. **Software quality assurance:** These are activities required to maintain software quality. Perform actions to ensure the product's quality.

4. **Formal technical reviews:** It is required to assess engineering work products to uncover and remove errors before they propagate to the next activity. At each level of the process, errors are evaluated and fixed.

5. **Software configuration management:** Managing of configuration process when any change in the software occurs.

6. **Work product preparation and production:** The activities to create models, documents, logs, forms, and lists are carried out.

7. **Reusability management:** It defines criteria for work product reuse. Reusable work items should be backed up, and reusable software components should be achieved.

8. **Measurement:** In this activity, the process can be defined and collected. Also, project and product measures are used to assist the software team in delivering the required software.

Personal and Team Process Models

Personal Software Process (PSP):

The Personal Software Process (PSP) is a structured software development process that is designed to help software engineers better understand and improve their performance by bringing discipline to the way they develop software and tracking their predicted and actual development of the code.

1. **Planning**: This activity isolates requirements and specifications to be decided prior to the development
   and estimates the size and the cost of the project. Defect estimation is also made. Development tasks
   are identified and scheduled is created.
2. **High-Level Design**: External specifications and requirements for each component to be constructed
   and developed. Component design is also created. Prototypes are developed if requirements are
   complex.
3. **High Level Design Review**: Formal verification methods are applied to uncover errors in the design.
4. **Development**: The component level design is reviewed and refined. The code is generated, reviewed,
   compiled and tested.
5. **Postmortem**: Using Metrics and measures the effectiveness of the process is determined. It provides
   guidance for modification of process and its improvement.

Team Software Process (TSP):

In combination with the personal software process (PSP), the **team software process** (TSP) provides a defined operational process framework that is designed to help teams of managers and engineers organize projects and produce software for products that range in size from small projects of several thousand lines of code.

1. Review project objectives and set team goals
2. Make a development plan for entire project.
3. Plan for the needed support facilities.
4. Make a quality plan and set quality targets.

5. It accelerate (increases) the software process development.
6. Define the team development process.
7. Merge the individual plan to team plan.
8. Establish team roles.
9. Rebalance team workload.
10. Access project risks and assign responsibilities.

Process Technology

Process technology refers to the tools that create, analyze, manage and improve processes in an organization.

The process technology tools allow a software organization to build, analyze their current process, organize work task, control and monitor progress and manage technical quality.

Process technology toolsgather and analyze process data to help organization to

- Determine their workflow and the people involved
- Examine alternative process structures
- Estimate the time the process requires
- Identify the obstacle in the process
- Lead to reduce development time or cost

Once the acceptable process has been created,other process technology tools can be used to allocate ,monitor and even control all software engineering tasks.

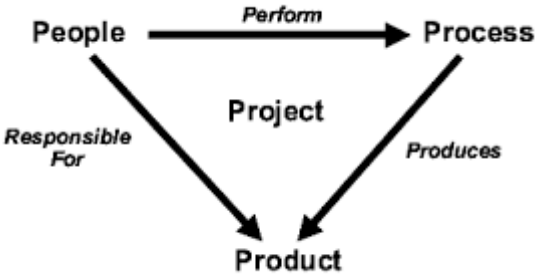
The process technology tool can also be used to coordinate the use of other computer aided software engineering tools for a particular work task.

Product and Process

In software engineering product refers to end result of the development process. On the other hand, the process refers to the steps and activities involved in creating product.

Product includes any software manufactured based on the customer's request. This can be a problem solving software or computer based system. It can also be said that this is the result of a project.

Process is a set of sequence steps that have to be followed to create a project. The main purpose of a process is to improve the quality of the project.



Product is the final production phase in the project. Process refers to a set of sequence of steps that should be followed with the goal of creating a project. Product focuses on the final result. Process focuses on completing every step involved in the project that is being developed.

It includes tasks like requirement gathering, design, coding, testing and deployment.

The process ensures that the product is developed efficiently, meets the desired output and delivered on time.

In simple words, the "**product**" is what you get and "**process**" is how you get it.

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